

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

7 Further Graphics (aco41)

Surface
representations

- (a) State the best geometry representation for the following tasks. Briefly explain.
- (i) Testing if a line segment intersects a surface. [1 mark]
 - (ii) Tracking the surface of a fluid undergoing geometry and topology (connectivity) changes. [1 mark]
 - (iii) Aligning and merging surfaces acquired from different views. [1 mark]
 - (iv) Rendering the surface of a cube. [1 mark]
 - (v) Look up the colour of each point on a surface from a texture. [1 mark]

Answer:

- (i) An implicit representation. It allows for inside/ outside/ on surface tests. [1 mark]
 - (ii) An implicit representation or a point cloud. Both allow for easy topology changes, in contrast with meshes and parametric surfaces. [1 mark]
 - (iii) An implicit representation or a point cloud. Meshes and parametric representations are difficult to merge. [1 mark]
 - (iv) A mesh. Polygons are efficient to both represent and render flat surfaces with sharp boundaries. [1 mark]
 - (v) A parametric surface or a mesh with uv-coordinates defined to provide a 2D parameter domain. [1 mark]
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Curvature for
closed surfaces

- (b) Assume constant minimum κ_{min} and maximum κ_{max} curvature for a surface. Which of the following could be a closed surface?
- (i) $\kappa_{min} = \kappa_{max}$. [1 mark]
 - (ii) $2H^2 - K = 0$ for mean curvature H and Gaussian curvature K . [2 marks]

Answer:

- (i) Yes, this can be closed surface, e.g. a sphere. [1 mark]
 - (ii) $H = (\kappa_{min} + \kappa_{max})/2$ and $K = \kappa_{min}\kappa_{max}$. $2H^2 - K = (\kappa_{min} + \kappa_{max})^2/2 - \kappa_{min}\kappa_{max} = (\kappa_{min}^2 + \kappa_{max}^2)/2 = 0$. This implies $\kappa_{min} = \kappa_{max} = 0$. [1 mark] This cannot be a closed surface, it describes a plane. [1 mark]
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Surface normal of
a heightfield

- (c) A heightfield is a surface in 3D defined by a function $h(x, y)$ over the xy -plane. Assuming we define points with $z > h(x, y)$ as outside this surface, provide an expression for the surface normal in terms of the derivatives of h without using the cross product. [4 marks]

Answer: We can define an implicit surface by $f(x, y, z) = h(x, y) - z = 0$. [1 mark] The surface normal is then defined by taking the derivatives of this implicit function, [1 mark]

$$f_x = h_x, f_y = h_y, f_z = -1$$

and defining the right sign [1 mark] and normalising. [1 mark]

$$\nabla f = [-h_x, -h_y, 1]/\sqrt{h_x^2 + h_y^2 + 1}$$

Surface normal of
a parametric/
implicit surface

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- (d) Given a surface in both parametric $\mathbf{p}(u, v)$ and implicit $f(\mathbf{p}) = 0$ forms,
- (i) prove that the surface normal of the parametric form is parallel to the normal of the implicit form.
[Hint: Chain rule: $\frac{\partial f(a(x), b(x), c(x))}{\partial x} = \frac{\partial f}{\partial a} \frac{\partial a}{\partial x} + \frac{\partial f}{\partial b} \frac{\partial b}{\partial x} + \frac{\partial f}{\partial c} \frac{\partial c}{\partial x}$] [6 marks]
 - (ii) Will the normals remain parallel if the implicit function is also a signed distance function? [1 mark]
 - (iii) Why is the sign ambiguous? [1 mark]

Answer:

- (i) We denote the implicit representation with $f(\mathbf{p}) = 0$ and parametric representation with $\mathbf{p}(u, v)$. It is possible to write: $f(\mathbf{p}(u, v)) = 0$. [1 mark] The value of the implicit function will not change if we change u or v as $\mathbf{p}(u, v)$ is always on the surface. [2 marks] Using this fact and the chain rule in the hint: $f_u = f_x x_u + f_y y_u + f_z z_u = \nabla f^T \mathbf{p}_u = 0$ and $f_v = f_x x_v + f_y y_v + f_z z_v = \nabla f^T \mathbf{p}_v = 0$. [2 marks] This implies that the gradient of f is orthogonal to the plane spanned by $\mathbf{p}_u$ and $\mathbf{p}_v$, hence is parallel to their cross product and parallel to the surface normal of the parametric form. [1 mark]
 - (ii) Yes, as long as it defines the same surface, i.e. the zero set is the same, we get the same result.
 - (iii) This is because for an implicit function, inside/ outside corresponding to positive and negative values can be arbitrarily defined.
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